

Flexible Bayesian Models for Statistical Demography

Herb Susmann

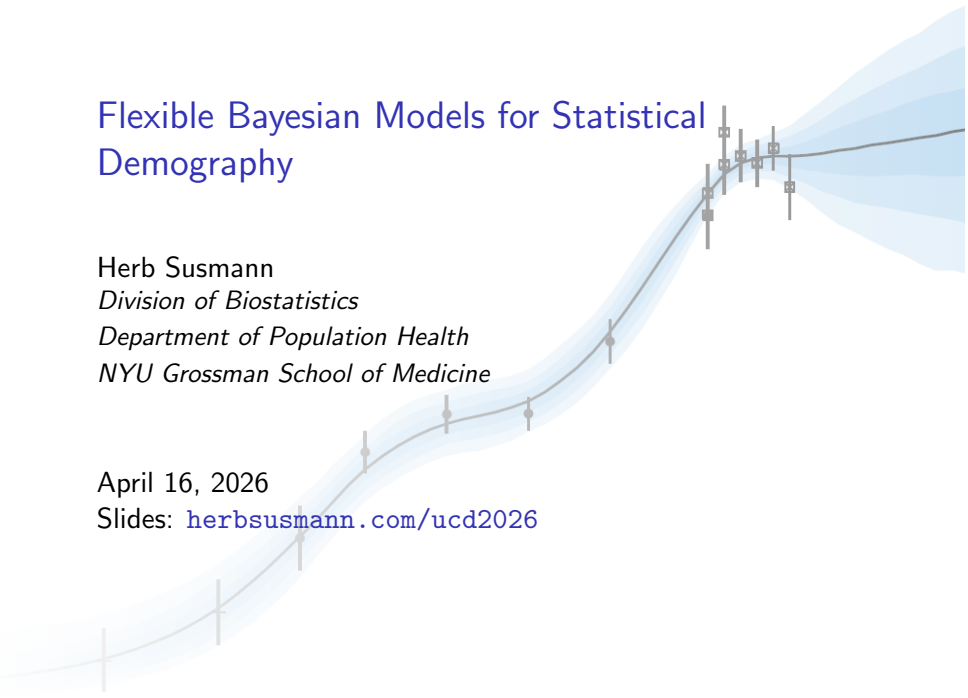
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Slides: herbsusmann.com/ucd2026



Research programme

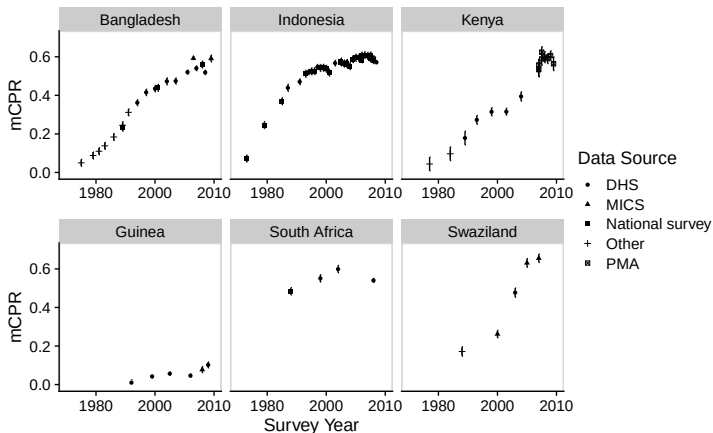
- ① *Bayesian modelling*: flexible hierarchical models for demographic estimation and forecasting.
- ② *Causal inference*: semiparametric, machine-learning-based methods for observational data.

Unifying agenda: Bayesian causal inference for statistical demography.

Research pillar: Bayesian modelling

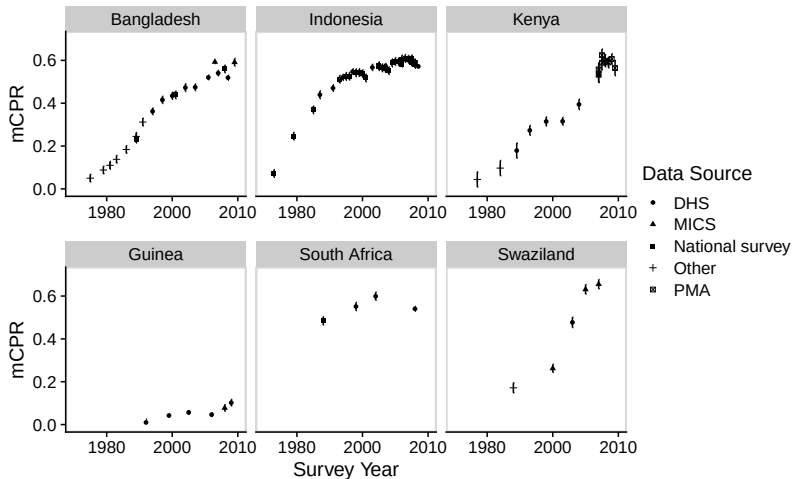
Demographic and health indicators track population changes over time.

Example: modern contraceptive use rate

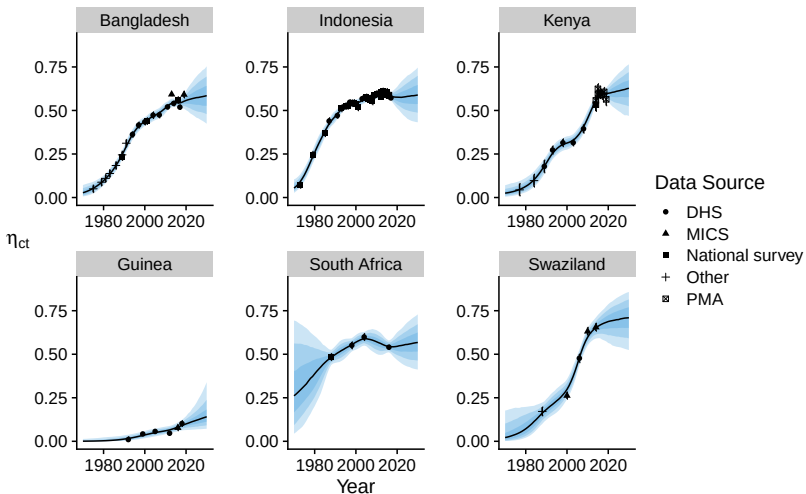


(Susmann and Alkema 2025 JRSS-C)

The challenge: from noisy, sparse data from multiple sources...



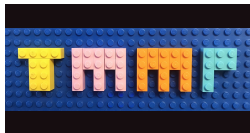
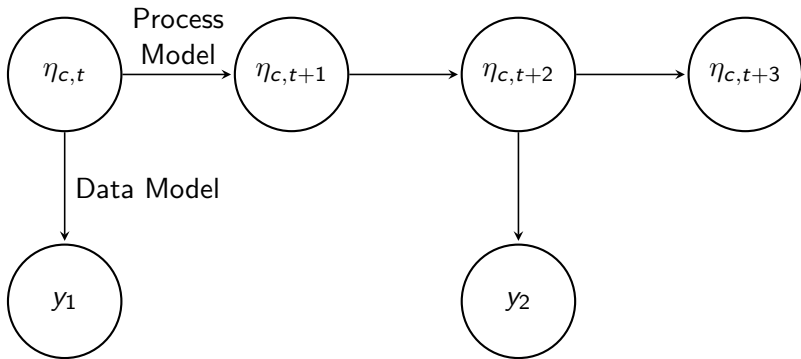
...produce probabilistic estimates and projections



Contribution: general modelling framework for demographic and health indicators

- **Challenge:** many statistical models have been created to provide estimates and projections, but...
 - comparing models can be difficult.
 - building a new model requires starting from scratch.
- **Contribution:** a modelling framework that decomposes models into interchangeable components, based on a common notation, that can be mixed and matched. (Susmann, Alexander, and Alkema, 2022 Int. Stat. Rev)

Temporal Models for Multiple Populations



Process Model

The **Process Model** defines how the true value of the indicator is expected to evolve over time.

$$g_1(\eta_{c,t}) = \underbrace{g_2(X_{c,t}, \beta_c)}_{\text{covariate}} + \underbrace{g_3(t, \eta_{c,s < t}, \alpha_c)}_{\text{systematic}} + \underbrace{\epsilon_{c,t}}_{\text{smoothing}}$$

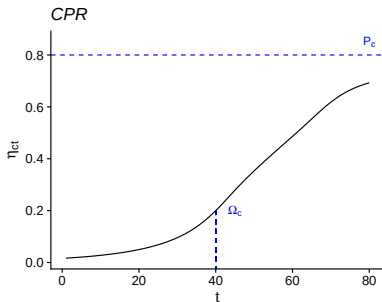
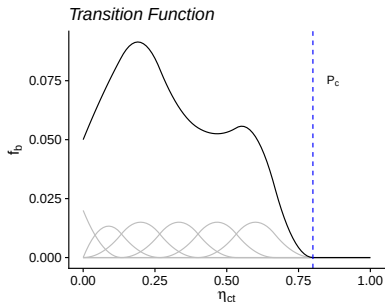
- **Covariate** → models associations with covariates
- **Systematic** → captures long-term expected trends.
- **Smoothing** → allows data-driven deviations from the other components, while still enforcing smoothness.

Contribution: modeling demographic transitions with B-splines

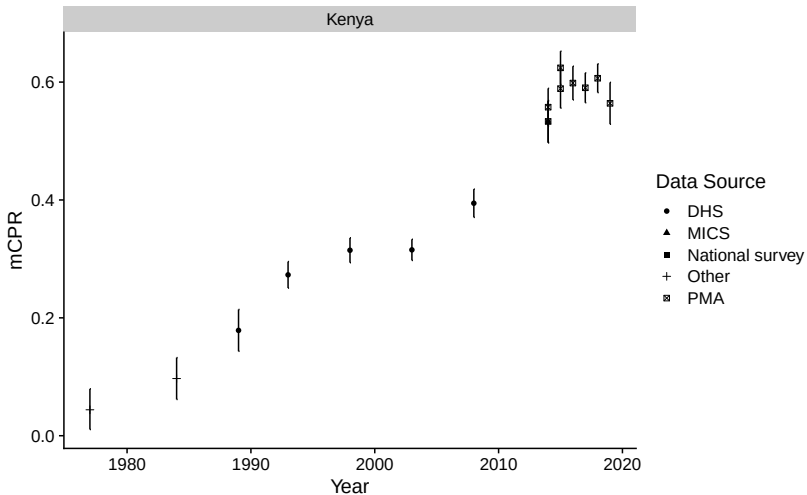
- **Challenge:** modeling indicators that follow *transitions* between stable states.
- **Contribution:** a general, flexible model applicable to a variety of indicators, without making strong functional form assumptions. (Susmann and Alkema, 2025 JRSS-C)
- **Key idea:** model the rate of change as a flexible B-spline function of the current level:

$$f_b(\eta_{c,t}, P_c, \beta_c) = \sum_{j=1}^J \underbrace{h_j(\beta_{c,j})}_{\text{coefficient}} \cdot \underbrace{B_j(\eta_{c,t}/P_c)}_{\text{basis function}}.$$

B-spline transition functions capture a large class of smooth transitions

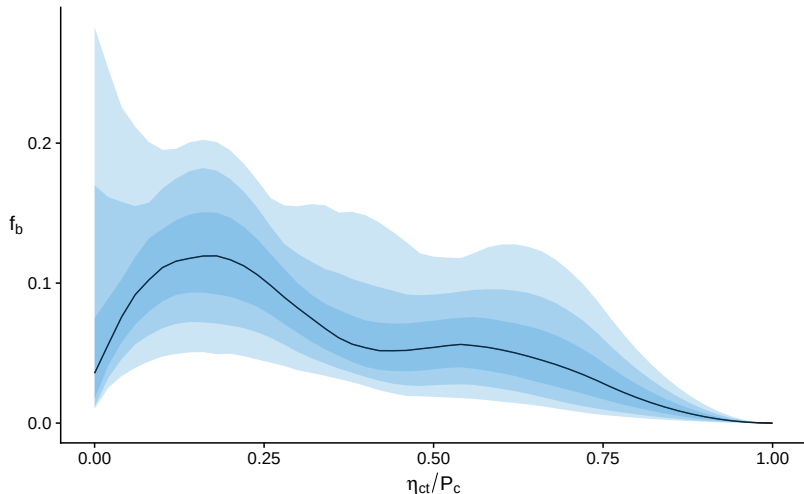


Kenya: raw data



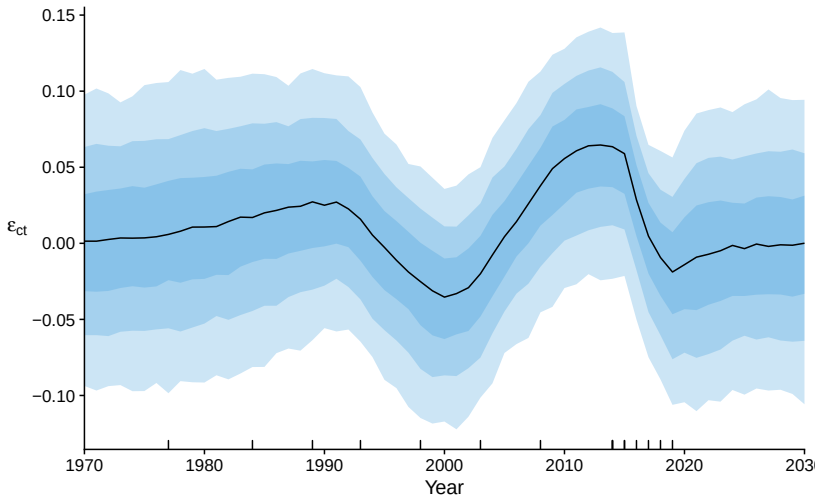
$$\text{logit}(\eta_{c,t}) = \underbrace{g_3(t, \eta_{c,s < t}, \alpha_c)}_{\text{systematic}} + \underbrace{\epsilon_{c,t}}_{\text{smoothing}}$$

Kenya: transition function



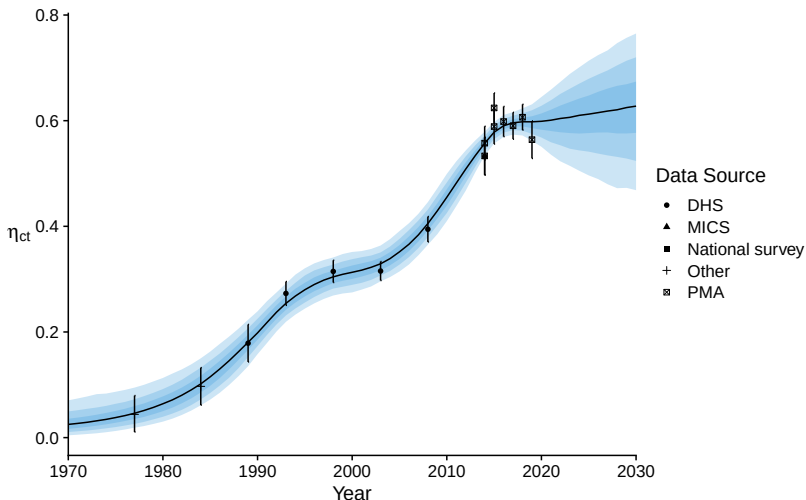
$$\text{logit}(\eta_{c,t}) = \underbrace{g_3(t, \eta_{c,s < t}, \alpha_c)}_{\text{systematic}} + \underbrace{\epsilon_{c,t}}_{\text{smoothing}}$$

Kenya: smoothing component



$$\text{logit}(\eta_{c,t}) = \underbrace{g_3(t, \eta_{c,s < t}, \alpha_c)}_{\text{systematic}} + \underbrace{\epsilon_{c,t}}_{\text{smoothing}}$$

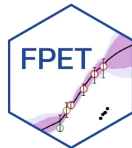
Kenya: Probabilistic estimates and projections



$$\text{logit}(\eta_{c,t}) = \underbrace{g_3(t, \eta_{c,s < t}, \alpha_c)}_{\text{systematic}} + \underbrace{\epsilon_{c,t}}_{\text{smoothing}}$$

Impact

- Presented at UN Population Division seminar (2025).
- B-spline transition model integrated in the *Family Planning Estimation Tool*: Used by countries to produce estimates of family planning indicators such as mCPR.
- Results used within countries and in the FP2030 global initiative.



FPET training by Avenir Health, Kenya 2024

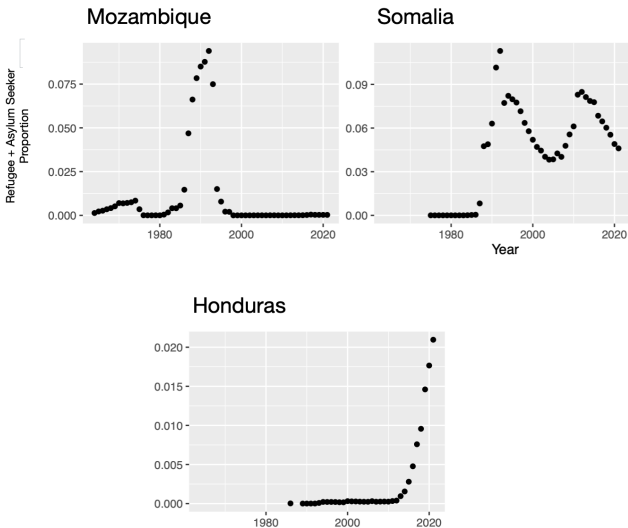
Contribution: modeling refugee and asylum seeker populations with Interrupted Logistic Models

- Migration plays a key role in population projections:

$$\text{Population}_{t+1} = \text{Population}_t + \text{Births}_t - \text{Deaths}_t + \text{Migration}_t$$

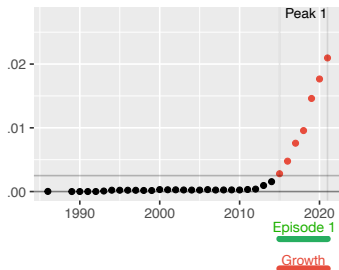
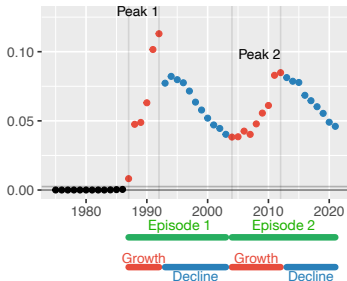
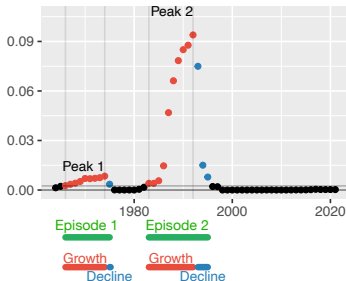
- **Challenge:** projecting refugee and asylum seeker populations is difficult due to highly contingent underlying processes.
- **Contribution:** a novel Bayesian model for refugee and asylum seeker populations. (Susmann and Raftery, 2025 Demography)

Refugee and asylum seeker population data

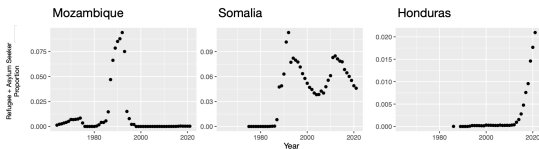
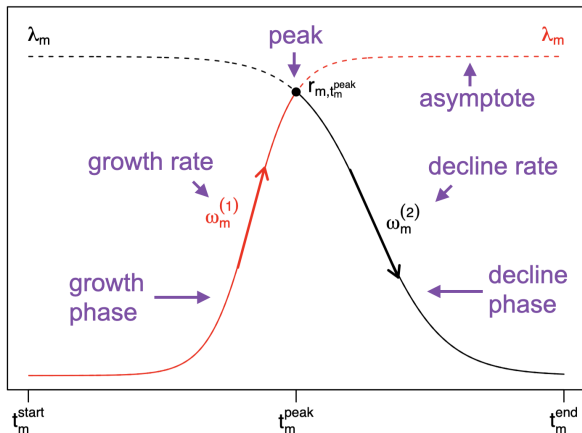


(Susmann and Raftery 2025 Demography; Data: United Nations High Commissioner for Refugees)

Segmentation into displacement episodes

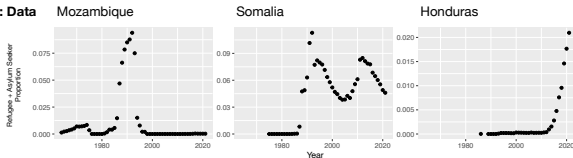


Interrupted Logistic Model: Structure

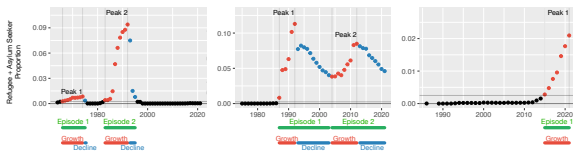


Integrated modeling pipeline

Step 1: Data

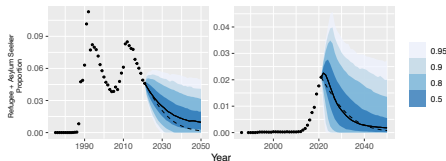


Step 2: Segmentation



Step 3: Estimation and Projection

All crises in Mozambique are classified as ended; no projections produced.



Research pillar: Causal inference

Core theme: robust causal inference for complex observational data.

- ① **High dimensional exposures** → Nonparametric shrinkage estimators
 - Asymptotically efficient estimation of high-dimensional causal parameters.
 - Susmann et al. 2026, Annals of Applied Statistics
- ② **Complex longitudinal exposures** → Generalised Longitudinal Modified Treatment Policies
 - Extending the Average Treatment Effect on the Treated to time-varying, multi-valued, and continuous exposures.
 - Susmann et al. 2024, under review, Biometrics
- ③ **Overlap violations** → Non-overlap bounds
 - Informative partial identification bounds when standard estimators fail.
 - Susmann, McClean, and Díaz 2025, preprint

Contribution: partial identification bounds for causal effects

Overlap assumption: every member of the population must have a positive probability of exposure.

Example: what is the causal effect of exposure to extreme drought on migration?

- Some people live in regions that never experience persistent drought.

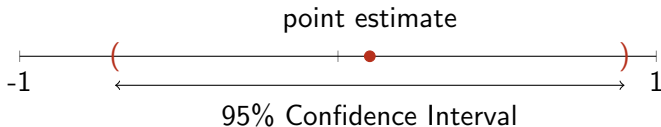
Key idea: (Susmann, McClean, and Díaz 2025, ArXiv preprint):
Decompose the Average Treatment Effect into overlap (identifiable) and non-overlap (non-identifiable) components:

$$\psi = \underbrace{\psi_O}_{\text{overlap}} + \underbrace{\psi_{NO}}_{\text{non-overlap}}$$

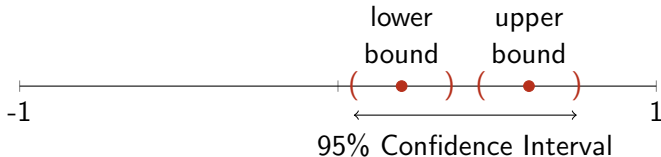
- ψ_O : estimated via doubly-robust methods; $\sqrt{n}$ -consistent and semiparametrically efficient under weak conditions.
- ψ_{NO} : can be bounded under minimal assumptions.

Our proposal: estimate *bounds* on the treatment effect

- Traditional approach: estimate **point estimate** and **95% confidence interval**:



- Our approach: estimate **lower bound** and **upper bound** and combine to form a **95% confidence interval**:

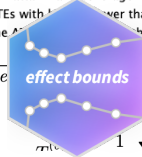


[Submitted on 24 Sep 2025]

Non-overlap Average Treatment Effect Bounds

Herbert P. Susmann, Alec McClean, Iván Díaz

The average treatment effect (ATE), the mean difference in potential outcomes under treatment and control, is a canonical causal effect. Overlap, which says that all subjects have non-zero probability of either treatment status, is necessary to identify and estimate the ATE. When overlap fails, the standard solution is to change the estimand, and target a trimmed effect in a subpopulation satisfying overlap; however, this no longer addresses the original goal of estimating the ATE. When the outcome is bounded, we demonstrate that this compromise is unnecessary. We derive non-overlap bounds: partial identification bounds on the ATE that do not require overlap. They are the sum of a trimmed effect within the overlap subpopulation and worst-case bounds on the ATE in the non-overlap subpopulation. Non-overlap bounds have width proportional to the size of the non-overlap subpopulation, making them informative when overlap violations are limited --- a common scenario in practice. Since the bounds are non-smooth functionals, we derive smooth approximations of them that contain the ATE but can be estimated using debiased estimators leveraging semiparametric efficiency theory. Specifically, we propose a Targeted Minimum Loss-Based estimator that is $\sqrt{n}$ -consistent and asymptotically normal under nonparametric assumptions on the propensity score and outcome regression. We then show how to obtain a uniformly valid confidence set across all trimming and smoothing parameters with the multiplier bootstrap. This allows researchers to consider many parameters, choose the tightest non-overlap bound estimator, and still attain valid coverage. We demonstrate via simulations that non-overlap bounds are more informative than traditional doubly-robust point estimators. We illustrate



2. For B bootstrap sample b , compute the studentized residual

$$r(b) = \frac{1}{\sqrt{n}} \sum_{i=1}^n r_i(b)$$

effect bounds $\mathbb{E}(\xi) = 0$, $\mathbb{E}(\xi^2) = 1$, and form

$$1 - \frac{1}{\sqrt{n}} \sum_{i=1}^n r_i(b) \left[\varphi_{U,k}(Z_i, \hat{\eta}) - \hat{U}_k \right]$$

Research vision: Bayesian modelling meets causal inference for statistical demography

- **Bayesian modelling:** How have populations changed, and how will they change?
- **Causal inference:** How *would* trends change under counterfactual interventions or policies?
- **Example:** Using large-scale survey data, estimate the causal effects of climate shocks on migration flows, and project how migration patterns would change under alternative climate and policy scenarios.

Summary

Two research pillars:

- ① **Bayesian modelling** for demographic estimation and projection
 - Modular framework (Int. Stat. Review), flexible B-spline transition models (JRSS-C), refugee modelling (Demography).
 - Methods adopted in Family Planning Estimation Tool; presented at UN Population Division.
- ② **Causal inference** for realistic observational data
 - Non-overlap bounds (preprint), generalised longitudinal effects (Biometrics, rev.), penalised estimation (AOAS).

Research vision: unify both pillars to advance Bayesian causal inference methodology for statistical demography.

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Transition Models

- A model class for indicators that follow a transition.
- *Transition Models* have a process model given by

$$g_1(\eta_{c,t}) = \underbrace{g_3(t, \eta_{c,s \leq t}, \alpha_c)}_{\text{systematic}} + \underbrace{\epsilon_{c,t}}_{\text{smoothing}}.$$

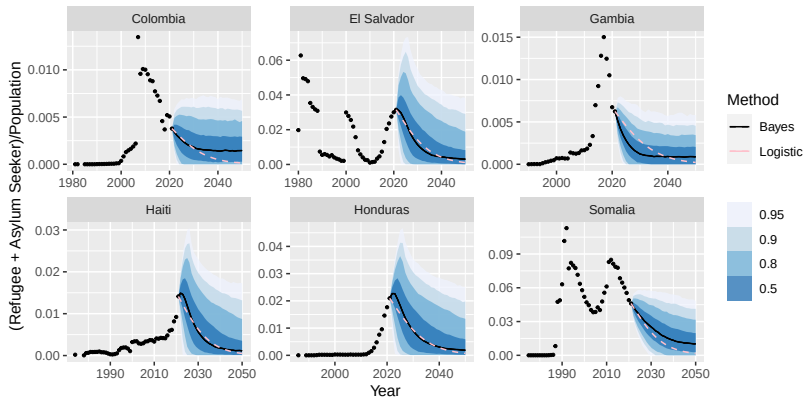
- The systematic component has the following form:

$$g_3(t, \eta_{c,s \neq t}, \alpha_c) = \begin{cases} \Omega_c, & t = t_c^*, \\ g_1(\eta_{c,t-1}) + f(\eta_{c,t-1}, P_c, \beta_c), & t > t_c^*, \\ g_1(\eta_{c,t+1}) - f(\eta_{c,t+1}, P_c, \beta_c), & t < t_c^*, \end{cases}$$

where $\alpha_c = \{\Omega_c, P_c, \beta_c\}$.

- The function f is called the *transition function*.

Projections for six illustrative countries



How the bounds are constructed

- Divide the population into **overlap** and **non-overlap** subpopulations.
- Set an *overlap threshold* c .
 - If $\pi \in [c, 1 - c]$ then unit is in the **overlap** subpopulation.
 - If $\pi \notin [c, 1 - c]$ then unit is in the **non-overlap** subpopulation.
- Assume that the outcome is *bounded*: $0 \leq Y \leq 1$.
- The ATE is bounded by:

$$\begin{aligned}\psi &= (Y^1 - Y^0) \\ &= (Y^1 - Y^0 \mid \pi \in [c, 1 - c])\mathbb{P}(\pi \in [c, 1 - c]) \\ &\quad + (Y^1 - Y^0 \mid \pi \notin [c, 1 - c])\mathbb{P}(\pi \notin [c, 1 - c]) \\ &\in (Y^1 - Y^0 \mid \pi \in [c, 1 - c])\mathbb{P}(\pi \in [c, 1 - c]) \\ &\quad \pm \mathbb{P}(\pi \notin [c, 1 - c]).\end{aligned}$$

Non-overlap bounds, formally

Let $\pi(X) = \mathbb{P}(A = 1 \mid X)$, and fix $c \in [0, \frac{1}{2}]$ denote a propensity score trimming threshold. Under consistency, no unmeasured confounding, and outcome boundedness,

$$(Y^1 - Y^0) \in [L(c), U(c)].$$

where

$$L(c) = \psi(c) - \mathbb{P}((X) \geq 1 - c),$$

$$U(c) = \psi(c) + \mathbb{P}((X) \leq c), \text{ and}$$

$$\psi(c) = \{\mu_1(X)\mathbb{I}(\pi(X) > c) - \mu_0(X)\mathbb{I}(\pi(X) < 1 - c)\}.$$

Practical benefits: higher precision, higher power

N	Mean Width		Power	
	Doubly-robust	Bounds	Doubly-robust	Bounds
<i>(A) Simulations conditional on non-overlap</i>				
100	1.91	0.46	0.02	0.49
250	1.83	0.29	0.07	0.88
500	1.65	0.22	0.16	1.00
1000	1.35	0.17	0.31	1.00